



Tandy® 1000 300 Baud Modem Board Owner's Manual and Installation Guide

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Important Note

Before connecting your Tandy 1000 Modem Board to the phone lines, notify your local telephone company of the:

Manufacturer: Radio Shack

Model: Tandy 1000

USOC Number: RJ11C

FCC Registration Number: AAO 88U-68616-DM-E

Ringer Equivalence Number (REN): 0.7B

***Tandy® 1000 300 Baud Modem Board
Owner's Manual and Installation Guide***

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Contents

INTRODUCTION	1
1 DESCRIPTION OF THE TANDY 1000 MODEM BOARD	3
2 SETTING UP THE TANDY 1000 MODEM BOARD ... Jumper Options	5
Installing the Board	5
Connecting the Telephone to the Board Modem	6
3 PROGRAMMING THE MODEM BOARD	7
4 TROUBLESHOOTING AND MAINTENANCE	9
5 SPECIFICATIONS	15
APPENDIX A How Quickly Does Your Modem Dial?	17
APPENDIX B Tandy 1000 Modem Board Programming Commands ..	19
APPENDIX C Technical Information	21
	25

Introduction

Congratulations on your purchase of the Tandy 1000 Modem Option Board. This modem has no switches or lights and is entirely controlled by the Tandy 1000 Computer keyboard entries. A card edge connector at one end of the card connects the modem to the computer's power circuitry. The other end of the card has two telephone connectors which let you attach a telephone in parallel with existing telephone circuitry. These ASCII commands are supplied through the on-board UART. The UART converts parallel commands to serial data for the modem. The microprocessor converts the serial data into parallel data and controls the modem using the input/output ports.

The modem circuitry is divided into six major subsections:

1. Modem Filter
2. Carrier Detect Section
3. UART
4. Microprocessor Section
5. Tone Dialer Section
6. Telephone Interface Section

Commands are used to program the Modem Board via the keyboard. You can program the Modem for Manual/Originate or Manual/Answer modes which means that if you are contacting a remote computer in Manual/Originate mode, you will manually dial the telephone number, and when you are in Manual/Answer mode, you will manually answer an incoming call. Or, you can program the Modem for Auto/Originate or Auto/Answer which means that when in Auto/Originate mode, the Modem will automatically dial the telephone number of a remote computer, and when in Auto/Answer mode, the Modem will automatically answer an incoming call.

1/Description of the Tandy 1000 Modem Board

- ① **Power Supply Connector.** Connects with the edge card connector socket located inside of the Tandy 1000.
- ② **Modular-to-Telephone Outlet Jack.** Connect one end of a modular-type cable to this jack. Connect the other end of the modular-type cable to the connector labeled, TO WALL. Connect the other end to a modular telephone outlet.
- ③ **Modem-to-Telephone Jack.** Connect the modular cable from the telephone to the jack labeled, TO PHONE.

Note: The TO WALL and TO PHONE jacks are interchangeable.

2/Setting Up the Modem Board

Jumper Options

The Modem board has built-in provision for configuring the board to match your requirements. It can be matched to the use of an external RS-232 and either a single-line (home type) telephone connection or the typical multi-line business installation.

On the board are 2 groups of electrical jumpers. A jumper is a small rectangular block of black plastic which has two openings in its base. When it is placed over 2 metal pins which extend about $\frac{1}{4}$ above the surface of the board, it electrically connects those pins together.

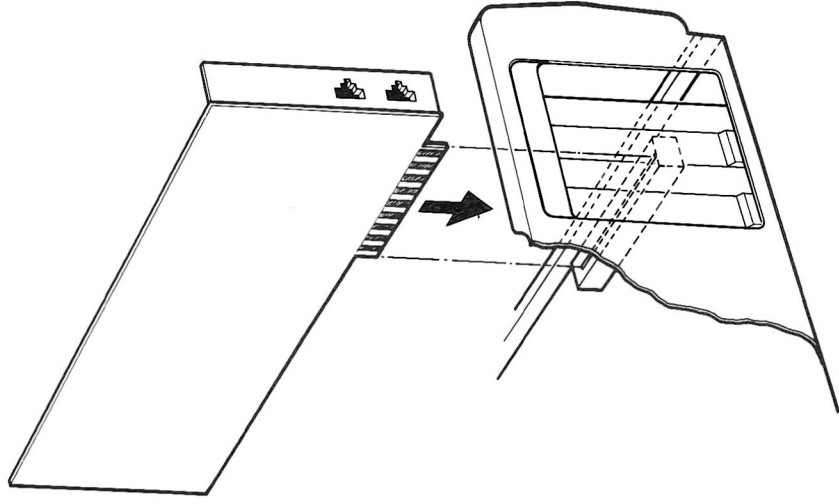


Figure 2. Installing the Modem Board

One set of 3 pins are labeled **PR**imary and **SE**condary (see Figure 2). As supplied, the Modem Option Board comes configured for the **PR**imary mode. If you plan to use only the Modem Option Board, you may skip to the Installation instruction page. However, if you already have in place or plan to immediately install the separate RS-232C Option Board, remove the jumper from the center pin and the pin marked **PR**I and, instead, place the jumper over the center pin and the one marked **SE**C.

Another set of 3 pins is designed to tailor the board to your phone system. If you use a standard home phone, place the jumper over pin **E**4 and **E**5. If you have a multi-line business phone, use pins **E**5 and **E**6.

Installing the Board

We highly recommend that you have this additional Board installed by the experienced professional technicians at your local Radio Shack/Tandy Computer Center. Doing so assures you that the finest in computer service techniques are followed, and that the appropriate tests and procedures are used.

If however, you **do** decide to install the board yourself, follow these steps exactly:

1. **Turn off all equipment.** If any unit is on, you can damage the central processing unit or the option board.
2. To avoid static electricity buildup, which can destroy certain integrated circuits, touch a grounded metal object before handling any circuit board. Do *not* stand on a carpeted floor while performing this work.
3. Remove the 2 screws on the front of the main unit. Remove the cover by sliding it away from the rear of the unit, parallel to the computer.
4. Rotate the chassis so that the fan faces you. To the immediate right of the fan are 3 option slot covers, each fastened to the rear of the chassis by a sheet-metal screw. On the Main Printed Circuit Board directly behind these covers, are 3 thin, black edge connectors. Select any empty socket as the location for the Board. It makes no difference which you use.
5. Remove the sheet-metal screw from the top of the selected cover. Tilt the metal strip away from you and lift the strip so that the tab on the panel clears the bottom of the slot. Store the cover in a safe place for possible future replacement.

6. Discharge any static buildup by touching a grounded metal object. Now grasp the Modem Option Board by its upper edges and position it above the black edge-connector socket. Insert the option card panel into the slot on the rear of the chassis, in the same way the option slot covers are mounted. At the same time, apply even downward pressure, engaging the edge-connector into its socket. Align the entire card so that the U-shaped cutout on the top of the option card panel is positioned over the screw-hole on the rear of the chassis. Replace the screw you removed before and tighten it to secure the board. Do not overtighten the screw.
7. Replace the top cover.

Connecting the Telephone to the Modem

The way you connect your Modem board depends on how your telephone is connected to the wall. The following figures illustrate these connections.

If your telephone has the older style, four-prong connection, use an adapter (Cat. No. 279-360) to connect it to your modem.

If your telephone is wired directly to the wall, contact your telephone company.

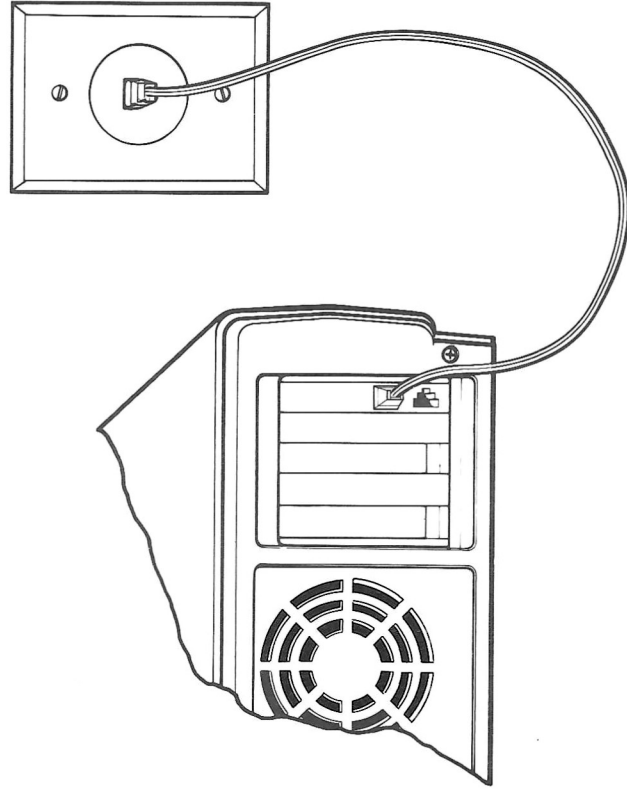


Figure 3. Modular Connector

3/Programming the Modem Board

The Tandy 1000 Modem board accepts data from the Tandy 1000 Computer, converts that data into tones and then sends the tones over telephone lines to another modem. It also accepts tones from telephone lines and changes those tones into data that your computer understands. You can program the Modem board to Auto Dial or Auto Answer and change other parameters simply by using the Programming Commands that let you communicate with the Modem.

The following criteria must be met before computer-to-computer communication can occur.

- Both computers must have the same communications protocol:
- Baud Rate (0-300)
- Word Length
- Parity
- Number of Stop Bits

See your Computer owner's manual to find out how to set communication protocol.

- One Modem must be in the ORIGINate Mode and the other must be in the ANSWer Mode. Usually, if you call another computer, you should be in the ORIGINate Mode. If you receive a call, you should be in the ANSWer Mode.

When you use your Modem in Automatic Mode, you don't have to dial any numbers or pick up your telephone receiver to answer a call—your Modem does all that for you. As long as you're going to be at your Computer throughout communications, you can control the Modem—and your telephone—through your communications programs such as Videotex.

You can also send or receive data from another computer while you're not at your Computer keyboard. Just write a BASIC or machine-language program and use the commands described in this manual (see Appendix B) to program your Modem to dial a number or answer the phone unattended. By programming your Modem to download daily receipts to another computer at 12:00 midnight, for example, you'll save time and be able to take advantage of lower telephone rates.

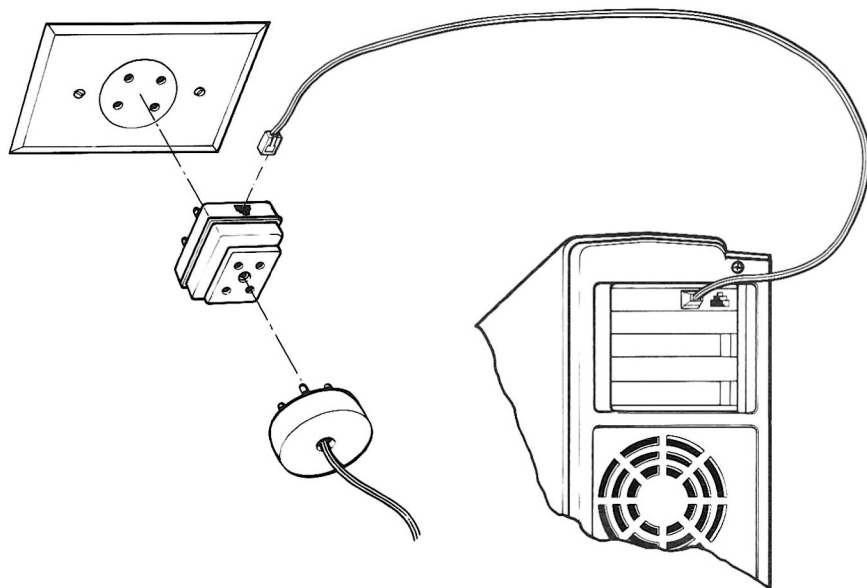


Figure 4. Wall Connector

A Brief Overview of the Modem Board's Commands

The Modem is completely programmable using the following commands:

Modem Board Programming Commands	
Command	Function
A	Switches modem to the Answer mode.
*	Opens modem for programming.
@	Stand by to receive a new Abort code.
C	Clears (resets) modem memory.
D	Clears current telephone number from memory. Must precede any new telephone number.
E	Toggle the Echo enable switch.
F	Fast Rotary or Tone dialing (20 pps.)
G	Toggle forced carrier detect switch.
L	Transmits (displays) current contents of modem memory to the computer (terminal).
M	Toggles the Manual/Automatic switch.
O	Switches the modem to the Originate mode.
P	Inserts a three-second pause in dialing sequence.
Q	Puts modem into LOCal Test Mode.
R	Rotary dialing (default value).
S	Slow Rotary or Tone dialing (10 pps.) (default value).
T	Tone dialing (10 digits per second).
X	Executes current program (dials phone number in memory, goes into Self-Test mode, waits for telephone to ring while in Answer mode).
[BKSP]	Erase the last code entered when programming dialing information.
(,) or -	Insert a user space when programming dialing information.

Table 1

Changing Communications Parameters

After you power-up the Tandy 1000 with the Modem board installed and load the communications software, you can change the default communications settings. You will be using some of the commands listed in Table 1. For a more thorough explanation of these commands, refer to Appendix B.

To see the current modem setting, use the List Command by typing **[*] [L]**.

The screen shows:

```
MODE=M
TYPE=O
F CD=Y
TEST=N
NUMB=
RATE=S
EXIT=14
```

Upon power-up, the modem will be set in the Manual/Originate mode, Forced Carrier Detect ON, no specified telephone number, slow dialing speed, and communications exit **[CTRL] [I]**. Each setting can be changed individually or by typing the settings in one line.

The following illustrates the possible settings:

MODE = M

The MODE setting can be either Manual or Automatic. If you select the Manual setting, the modem will not Auto Dial or Auto Answer. To change from Manual to Automatic type **[M] [L]**. To toggle back to Manual mode, type **[M] [L]**. When you press **[L]**, the current parameter settings are displayed.

TYPE = O

Using the TYPE setting, you can place the Modem in either Originate or Answer Mode. When you are calling another computer, you will need to place the Modem in the Originate setting. If the MODE option is A and the TYPE option is O, the Modem board will Auto Dial the number you have saved in its memory (see NUMB=).

When you will be receiving a call, you use the Answer option (A). If the MODE option is A and the TYPE option is A, the Modem will wait for the telephone to ring, then answer the call. If the MODE option is M and the TYPE option is A, the Modem will answer the phone as soon as you type X.

To change the TYPE setting to Originate, type **[O] [L]**. To change the TYPE setting to Answer, type **[A] [L]**.

F CD = Y

The F CD line determines whether or not the modem will supply the computer with an asserted Carrier Detect. Upon system initialization, the F CD line will show Y, meaning yes. If you are running software that does not permit serial communications unless Carrier Detect is ON, be cautious when using this command. If you indicate that Carrier Detect is OFF, you will have no way to turn it back on without turning the Tandy 1000 Off, then On again.

Type **G L** to tell the Modem to supply the computer with an asserted Carrier Detect ONLY when the Modem is in communications over the telephone lines. F CD = N will display when the List command is used.

Type **G** to tell the modem to supply the computer with an asserted Carrier Detect, even though you are NOT in communications over the telephone lines.

TEST = N

This lets you turn the Self Test Mode ON or OFF. When the Self Test Mode is ON (TEST = Y), the modem will NOT connect with telephone lines but will let you test to see if typed characters are displayed on the screen after the execute command (S) is used.

To change the Self Test Mode, type **G L**.

NUMB =

If you are in Automatic/Originate mode, you can program the modem to automatically dial the telephone number. At power-up, NUMB = has no predesignated telephone number.

To enter a telephone number, first press **D**, type the telephone number, and then press **L**. The telephone number will display with the other current parameter settings.

In addition to the telephone number, you can also specify whether you are using a Tone telephone (T) or a Rotary telephone (R), and any pauses during dialing (P).

For example, the sequence **DT9P1234567** tells the modem that you are using a Tone telephone, to dial 9, pause 2 seconds, and then dial the number 1234567.

RATE = S

You can select either Fast or Slow dialing. The modem will auto dial a telephone number (if previously entered for NUMB =) using slow or fast speed Pulse or Tone signaling.

To change the dialing speed to Slow, type **S**.

To change the dialing speed to Fast, type **F**.

You can set all parameters in a single line. Note the following example:

MOGFDTP-1-(817)-555-1212

This example illustrates a program that would place the Modem in Auto/Originate mode. Also, Forced Carrier Detect is Off. The Modem is to Dial the following number using Tone at Fast Speed, then Dial 9, pause for three seconds, and then dial the rest of the number.

To reiterate, to execute the commands you programmed, press **X**.

4/Troubleshooting and Maintenance

If you have trouble transmitting data (garbled data, intermittent errors, etc.), be sure that:

- The phone connection is clean and noise-free.
- No one is talking on the telephone line.
- The phone and all extensions are on-the-hook.
- The baud rate of the terminals is equal to 300.

If you still can't locate the source of your problem, run the LOCal test.

Error Messages

- ? Your Modem has received a character which isn't in proper order or which it doesn't understand.

5/Specifications

Receive Frequencies

- A. Answer
Mark 1270 Hz
Space 1070 Hz
- B. Originate
Mark 2225 Hz
Space 2025 Hz

Transmit Frequencies

- A. Answer
Mark 2225 Hz
Space 2025 Hz
- B. Originate
Mark 1270 Hz
Space 1070 Hz

Transmit Level

- 13 dBm $\pm$ 1 dB
- 40 dBm
- 0-300

Receive Sensitivity

Baud Rate

Operating Temperature

55°F-85°F
(13°C-29°C)

APPENDIX A/ How Quickly Does Your Modem Dial?

You have the flexibility of controlling the dialing speed when using the Auto Dial feature of the Modem board. There are two types of dialing—Pulse (Rotary) and Tone. Each type can be specified as either Fast or Slow. The dialing speeds are applicable only when you are using the Modem in the Automatic/Originate Mode. The information following shows the speeds at which you can control dialing:

S Slow

In Slow speed, the Modem will Pulse dial at 10 pulses per second and will Tone dial at 5 characters per second. Either Pulse or Tone dial is significantly faster than you could dial a telephone manually. The Slow dialing speeds are shown in Table 2.

Dialing Using Slow Speed	
Pulse:	Break Time
	Make Time
	Intra Digit Time
Tone:	Tone On Time
	Tone Off Time

Table 2

F Fast

In Fast speed, the Modem will Pulse dial at 20 pulses per second and will Tone dial at 10 characters per second. The following table illustrates the capacities of Fast dialing. The Fast dialing speeds are shown in Table 3.

Dialing Using Fast Speed	
Pulse:	Break Time
	Make Time
	Intra Digit Time
Tone:	Tone On Time
	Tone Off Time

Table 3

APPENDIX B/ Modem Programming Commands

The following is a list of commands that you will be using to program or communicate with the Modem via keyboard:

*

Opens the Modem for Programming

This command forces the modem from an Off-Line status into the Command Receptive state which prepares the modem to accept commands from the computer. The * can be used in several places in the modem protocol. At initial power-up, you will need to precede any command with an *.

If used while the modem is waiting for a telephone call (waiting for the phone to ring), the * terminates the Answer mode and will not be echoed, but the TYPE option will be toggled to Manual. This ensures that the Modem is not accidentally told to go back to the WAIT FOR CALL mode.

If the * command is used while the Modem is dialing in Auto/Originate Mode, the * will not be echoed but the Modem will stop dialing and return to the Off-Line mode. To return the Modem to the command receptive state, you will need to type another * (which will be echoed).

The * command can be used at any time while in the command receptive state, it will be echoed to the computer, but no action will occur.

@

Abort On-Line Condition Character Programmer

This command lets you change the Abort On-Line code. Upon initialization, the EXIT code will be 14 HEX (CTRL T). You can change the code to any seven bit code. This command can be used to program the Modem to respond to an Abort command other than CTRL T. Any ASCII character may be substituted following the @. For example, to change the Abort code from CTRL T to CTRL R after doing a List command, you would type:

@ CTRL R

The Modem will watch for keyboard activity for two seconds after the Abort code is sent. If there is no activity during that two second period, the computer will disconnect the telephone line. If there is keyboard activity within that two second period, the Modem will ignore the Abort command.

The **CTRL** **(R)** would be listed as **EXIT = 12** (in **HEX**). (When you enter the **ASCII** character, it will not be echoed on the screen. Only the **@** will be echoed, provided the Echo switch is not disabled.)

C

Clear (reset) memory

The **C** command clears the Modem's memory and resets the Modem to its default values. There is no need to follow the **C** command with an execute command (**X**) because the memory is cleared immediately when the **C** is pressed.

D

Program the Following Telephone Number

The **D** command tells the modem to store the telephone dialing instruction within memory. The modem's memory can handle up to 25 digits of data at one time. Only those codes listed will be saved or used. The following codes are valid Dialing instructions:

0-9 Telephone number.

P **P** inserts a three second pause in the dialing sequence. This command is especially useful when you're dialing from a business system that requires an outside access code. For longer pauses, use more than one **P**. There is no limit as to the number of pauses that can be inserted.

R or T Used to indicate Rotary type dialing (pulse signalling) or Tone type dialing, respectively.

() Insert to separate portion of telephone number such as a prefix.

(BKSP) Delete last character entered.

-, (,) Insert to separate portion of telephone number.

The **T**, **P**, **R** and digits **0-9** can be in any position following the **D** command. Unless indicated, the Modem will dial using Pulse signalling. Therefore, if you want to dial with Tone, you must precede the telephone number with the **T** command.

You can mix signalling types in a number. For example, the sequence **D234333PPT1-(214)-988-1234** says to dial the number 234333 using Pulse signalling, pause for 6 seconds, and then dial the number 2149881234 using Tone signalling. The dashes and parentheses are ignored by the modem when dialing but are available for your use to make the sequence easier to read.

The **(BKSP)** code is simply an editing feature. If you are typing a number and make an error, press **(BKSP)** to delete the last character typed. If you press **(BKSP)** immediately after selecting the **D** command, a **?** mark will appear, indicating an error condition.

E

Echo On or Off

You can determine whether or not characters display on the screen using the Echo (**E**) command. The default parameter is an Echo enable. To toggle the Echo Enable/Disable, press **(E)**. When echo is disabled, the Modem will still receive commands but nothing will be displayed on your screen.

F

Fast Rotary or Tone dialing

If you do not specify how you want your Modem to dial the number, it will dial rotary-style at 10 pulses per second. This command changes the dialing speed to 20 pulses per second. Before using this command, contact your local telephone business office to check whether your telephone exchange can use 20 pps dialing.

Use this command to change the Tone dialing speed to 10 characters per second.

G

Toggle Forced Carrier Detect

The **G** command turns the Forced Carrier Detect **ON** and **OFF**. It is defaulted to turn **ON**. When **ON**, the Modem will present a false carrier command to the computer. If the **G** command is issued from a power-on condition, Carrier Detect will turn **OFF**. It will stay off unless a true carrier is asserted to the modem, due to On-Line status. To check the Carrier status, use the **List** command.

L

List options

This frequently used command displays the Modem's current parameter settings of Manual or Automatic, Answer or Originate, Forced Carrier Detect **On** or **Off**, Self-Test **On** or **Off**, Telephone Number, Dialing Speed, and Abort On-Line code. The **L** command may also follow the entry of new parameters.

Q *Toggle the Self-Test ON or OFF*

The Self-Test mode is useful to determine if the Modem is operating correctly. Upon initialization, the Self-Test mode is Off. When the Self-Test mode is turned on using the Q command, the Modem will not go On-Line if an X command is sent. Instead, the internal circuits will be adjusted to allow the Modem to "talk to itself." Always use the List command to check the status of the Self-Test mode.

S *Slow Ready or Tone Dialing*

Use this command to program the Modem to Rotary or Tone dial at 10 pulses per second.

X *Execute current program*

Send an X to immediately execute the program currently in the Modem's memory. At this point, the internal switches are read and the modem's response depends on your parameter designations.

APPENDIX C/ Technical Information

Monitor Mode

After the Modem establishes communications with a remote modem, it is considered to be in the Monitor Mode. While in the Monitor Mode, parameters are constantly scanned. If any one of these parameters fail, the modem will break communications and will go Off-Line immediately. The following parameters are monitored:

- **Carrier Detect**

The carrier monitored is that which is sent over the telephone line and of the false carrier that may be generated by the modem card. If this carrier is removed for longer than 300 mSeconds (.3 seconds), Monitor Mode is exited. If the carrier interruption is less than 300 mSeconds, the loss of carrier is ignored.

- **Manual Abort Entry**

If you send a **CTRL** (14 HEX), or another character that was programmed with the @ key, and do not send any other character for two seconds, the Modem card will exit the Monitor Mode. The DTR is also scanned in this mode. If the DTR is Off, the Modem will terminate communications, just as if the carrier had failed.

- **Modem Transmits a Space**

If you transmit a long space (a space which is 1.5 seconds or longer), the Modem will abort communications.

- **Modem Receives a Space**

If the Modem card receives a space from the remote modem that is longer than three seconds, communications will be aborted.

Modem Handshaking

End of Dial Code

In the Auto/Originate Mode, after the Modem has completed signaling the phone line, the Modem will issue a special code through the on-board UART. An ACK (06 HEX) is sent just after the modem has finished dialing the telephone number. In applications where the Modem is used as a directory dialer, not as a communications interface, this code can flag special software. The software can then take the Modem On-Hook, so that you may communicate with the dialed number using a telephone.

Communications Established Code

If the Modem receives a valid carrier detect from the telephone lines, it will issue a code string to the computer. When the computer receives a line feed, line feed, carriage return (0A 0A 0D HEX), the communications channel is ready. The three byte code sequences will operate at any time (in all modes of operation) the modem receives true carrier over the telephone lines.

Communications Failed Code

A code sequence is sent when the Modem fails to establish communications, or when it escapes the Monitor Mode to the Off-Line Mode. This sequence is:

NAK, line feed, carriage return (15 0A 0D HEX).

This sequence is valid in all operating modes.

Auto Originate with 'No Number To Dial' Detect

If you have failed to program at least one valid character into the Dialer memory, the Modem will abort the On-Line mode two seconds after the X command is issued. If there is a character or space programmed into the dialer memory by accident, the Modem will assume that that is the number and proceed to dial (if the character can be dialed). The Modem will then wait for 25.6 seconds before aborting the unsuccessful dial attempt.

Communications Examples

The following illustrates some of the methods you can use while in either the Manual or Automatic and/or the Originate or Answer modes.

1. With the Tandy 1000 and the communications software initialized, to use the default parameter settings, type: *X. When the *X command is used, the Modem will look for the carrier detect. If a number has not been dialed, the Modem will hang up after not locating a true carrier.

The Manual/Originate mode is useful when receiving computer files (such as programs) from a remote computer using communications software.

2. At power-up, to change parameters to Auto Dial so that the telephone number 1234567 will automatically be dialed, press [L] for the List command. Then type the following:

***MD1234567X**

3. With the system fully initialized, to indicate Fast Tone dialing of the above telephone number, you would type:

***MFD1234567X**

4. To change the Modem setting to Answer, after disconnect type:

***AX**

5. To change to Originate Mode, and dial the telephone number as indicated in #4 above, type:

*** (to turn off WAITING FOR RING)**

***OX**

6. After disconnect from #5, go to Manual Answer by typing:

***MAX**

7. After disconnect from #6, go to Manual Originate by typing:

***OX**

8. After disconnect from #7, go to Auto Answer by typing:

***MAX**

SERVICE POLICY

Radio Shack's nationwide network of service facilities provides quick, convenient, and reliable repair services for all of its computer products. In most instances, Warranty service will be performed in accordance with Radio Shack's Limited Warranty. Non-warranty service will be provided at reasonable parts and labor costs.

Because of the sensitivity of computer equipment, and the problems which can result from improper servicing, the following limitations also apply to the services offered by Radio Shack:

1. If any of the warranty seals on any Radio Shack computer products are broken, Radio Shack reserves the right to refuse to service the equipment or to void any remaining warranty on the equipment.
2. If any Radio Shack computer equipment has been modified so that it is not within manufacturer's specifications, including, but not limited to, the installation of any non-Radio Shack parts, components, or replacement boards, then Radio Shack reserves the right to refuse to service the equipment, void any remaining warranty, remove and replace any non-Radio Shack part found in the equipment, and perform whatever modifications are necessary to return the equipment to original factory manufacturer's specifications.
3. The cost for the labor and parts required to return the Radio Shack computer equipment to original manufacturer's specifications will be charged to the customer in addition to the normal repair charge.